

Unit 8, Lesson 3: Special Parallelograms



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can prove rectangles are parallelograms.
- ☐ I can use properties of parallelograms to identify special parallelograms.



8.3.1 Warm-Up: Estimation

1. A picture of a bowl with a cereal square is shown.



- a. Estimate the number of cereal squares that will go around the inner circumference of the bowl.

2. A picture of the bowl with a diameter of cereal squares is shown.

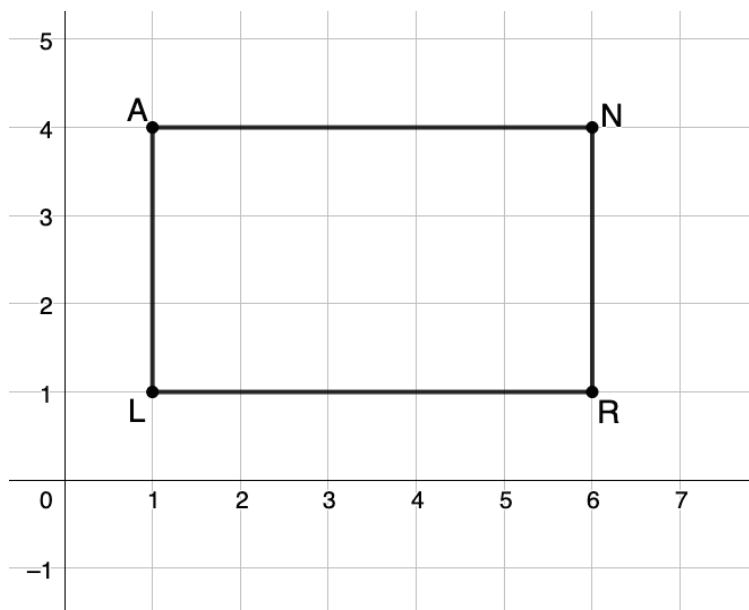


- a. Use the diameter of cereal squares to find the circumference.
- b. Compare your estimate to the circumference found using the diameter of cereal squares.



8.3.2 Collaboration: What is a Rectangle?

1. $ANRL$ is a rectangle shown on the coordinate grid.



- a. List everything you know about this type of quadrilateral.
- b. Compare your list with your partner's list. Add anything you may have missed.

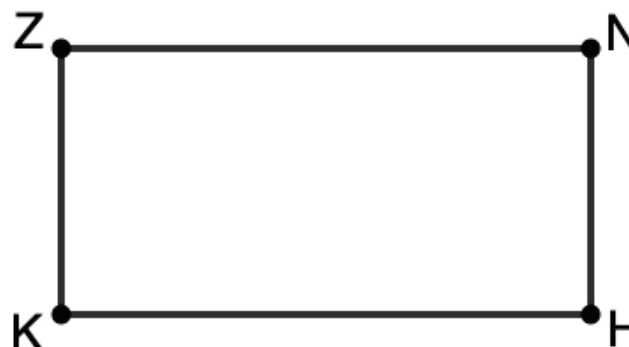
- c. With your partner, list two truths and one lie about this type of quadrilateral in the table below. Share your list with another pair. See if they can spot the lie.

1.

2.

3.

2. $ZNHK$ is shown.



- a. Using a ruler, draw $\overline{ZH}$ and $\overline{KN}$.
- b. These segments are called _____.

- c. Measure the length of these segments and write them in the table below. Include units.

Segment	Length
$\overline{ZH}$	
$\overline{KN}$	

- d. Make a conjecture about the lengths of the diagonals of a rectangle.
- e. Label the intersection point of the diagonals S .
- f. Measure the following segments and fill in their length in the table below. Include units.

Segment	Length
$\overline{ZS}$	
$\overline{HS}$	
$\overline{KS}$	
$\overline{NS}$	

- g. What do you notice?
- h. Make a conjecture about what happens to the diagonals of a rectangle when they intersect.



8.3.3 Exploration: Comparing Parallelograms and Rectangles

1. Work with your partner to fill in the table. If the statement describes a property always true of the shape, mark the box.

Statement	Parallelogram	Rectangle
opposite sides congruent	☐	☐
opposite sides parallel	☐	☐
opposite angles congruent	☐	☐
consecutive angles supplementary	☐	☐
diagonals bisect each other	☐	☐
diagonals are congruent	☐	☐
all angles are congruent	☐	☐

Based on the chart, discuss with your partner whether a rectangle could be classified as a parallelogram.



A **rectangle** is a quadrilateral that contains four right angles. Rectangles may be oblong or square. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.

2. $RPQD$ is a rectangle.



Complete the statements.

It is given that $RPQD$ is a rectangle. By the definition of

a rectangle $\angle R, \angle P, \angle Q$, and $\angle D$ are all ☐ right angles,
☐ acute angles,
☐ obtuse angles,

which means that $m\angle R = m\angle P = m\angle Q = m\angle D =$ ☐ 90° .
☐ 180° .

Since $m\angle P + m\angle Q = 180^\circ$ we can show that ☐ $\overline{RD} \parallel \overline{PQ}$
☐ $\overline{RP} \parallel \overline{DQ}$

and since $m\angle D + m\angle Q = 180^\circ$ we can show that ☐ $\overline{RD} \parallel \overline{PQ}$
☐ $\overline{RP} \parallel \overline{DQ}$

by the converse of the ☐ alternate interior angles
☐ consecutive interior angles

theorem. Therefore, $RPQD$ is a ☐ parallelogram
☐ kite

by definition.



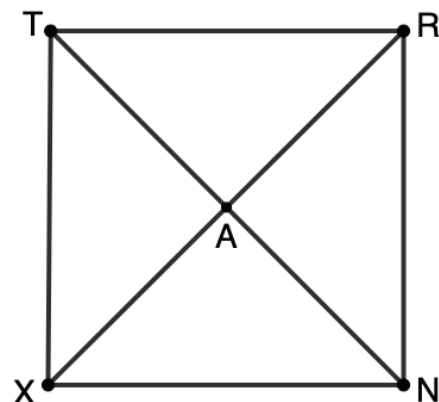
8.3.4 Guided Instruction: More Special Parallelograms



A **square** is a quadrilateral with four right angles and four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms, rectangles, rhombi.

1. $TRNX$ is a square with diagonals $\overline{TN}$ and $\overline{XR}$ intersecting at point A .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include the units.



Segment	Measure	Segment/Angle	Measure
$\overline{TN}$		$\overline{XA}$	
$\overline{XR}$		$\overline{RA}$	
$\overline{TA}$		$\angle XAT$	
$\overline{NA}$		$\angle NAR$	

- b. Which of the following describe the diagonals of a square? Circle all that apply.

congruent

bisect

perpendicular

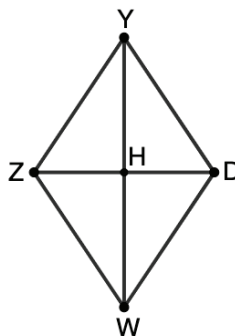
- c. List the features that determine if a parallelogram is a square.



A **rhombus** is a quadrilateral containing four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.

2. $YDWZ$ is a rhombus with diagonals $\overline{YW}$ and $\overline{ZD}$ intersecting at point H .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include units.



Segment	Measure	Segment/Angle	Measure
$\overline{YW}$		$\overline{ZH}$	
$\overline{ZD}$		$\overline{DH}$	
$\overline{YH}$		$\angle ZHW$	
$\overline{WH}$		$\angle YHD$	

- b. Which of the following describe the diagonals of a rhombus? Select all that apply.

congruent

bisect

perpendicular

- c. List the features that determine if a parallelogram is a rhombus.

- d. What are the differences between a square and a rhombus?

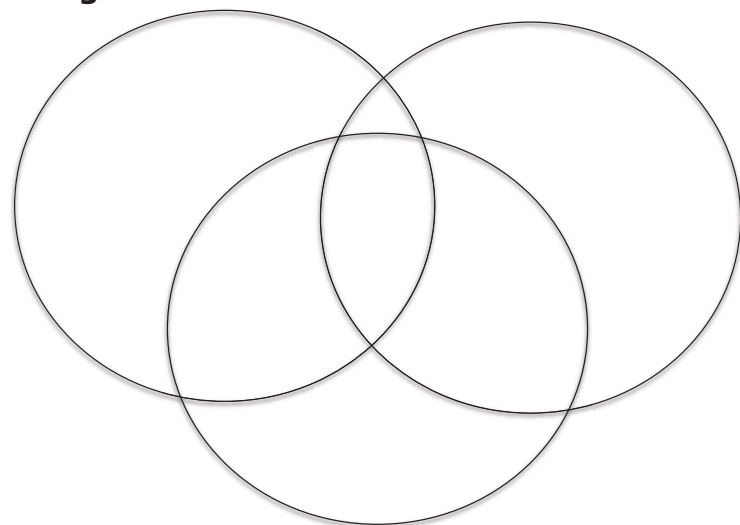


3. Use the numbered statements in the box to fill in the Venn diagram comparing rectangles, squares, and rhombi.

- | | |
|----------------------------------|---|
| 1. opposite sides are congruent | 6. consecutive angles are supplementary |
| 2. opposite sides are parallel | 7. all sides are congruent |
| 3. all angles are congruent | 8. diagonals are congruent |
| 4. diagonals are perpendicular | 9. diagonals bisect each other |
| 5. opposite angles are congruent | |

Rectangle

Rhombus



Square



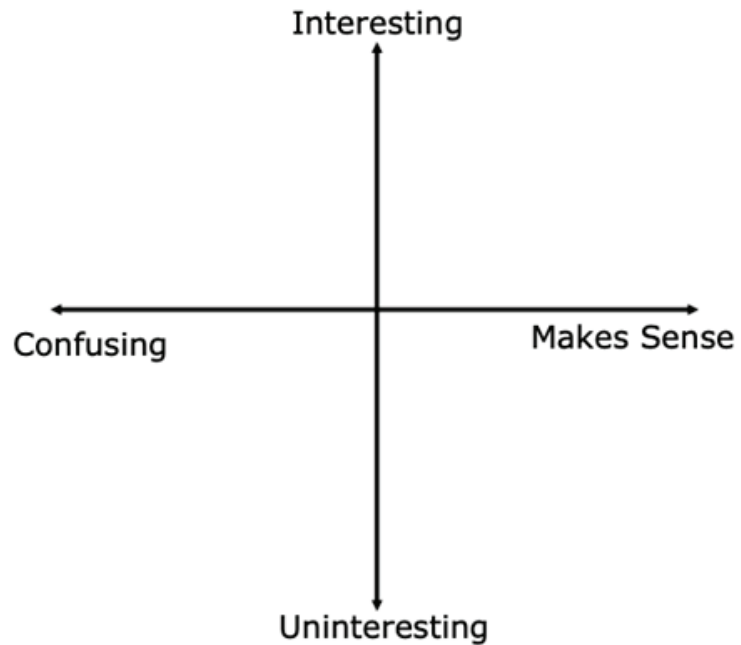
8.3.5 Bring It Together: Which Special Parallelogram Is It?

1. Explain how to show a parallelogram is a rectangle but not a rhombus or a square.
2. Explain how to show a parallelogram is a rhombus but not a square or a rectangle.
3. Explain how to show a parallelogram is a square.



8.3.6 Wrap-Up: Reflection

1. Plot a point on the grid to indicate your level of interest and how much sense you made of today's learning targets.



2. Explain why you placed the point where you did.



Unit 8, Lesson 4: Kites and Parallelograms



Benchmark

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.



Learning Target

- ☐ I can use properties of parallelograms, kites, and special parallelograms to solve mathematical and real-world problems.